

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{1}{8} g_1 g_2 \left(-2 c_Y^2 \overline{y_e^{pr}} y_e^{pr} + \sum_p c_{2Y} g_2^2 \right) LF_{3,0} [m_l^p] + \frac{1}{8} g_1 g_2 \left(-2 c_Y^2 \overline{y_e^{pr}} y_e^{pr} + \sum_p c_{2Y} g_2^2 \right) \right. \\ & LF_{4,-1} [m_l^p] + \frac{1}{8} g_1 g_2 \left(-2 c_Y^2 \overline{y_d^{pr}} y_d^{pr} + 2 s_Y^2 \overline{y_u^{pr}} y_u^{pr} + \sum_p c_{2Y} g_2^2 \right) LF_{3,0} [m_q^p] - \\ & \frac{1}{8} g_1 g_2 \left(-2 c_Y^2 \overline{y_d^{pr}} y_d^{pr} + 2 s_Y^2 \overline{y_u^{pr}} y_u^{pr} + \sum_p c_{2Y} g_2^2 \right) LF_{4,-1} [m_q^p] + \\ & \frac{1}{32} \left(g_2 g_1^3 (-1 + c_{4Y}) + g_1 g_2^3 (3 + c_{4Y}) \right) LF_{3,0} [m_\Phi] - \frac{1}{32} g_1 g_2 \left(g_1^2 (-1 + c_{4Y}) + g_2^2 (3 + c_{4Y}) \right) \\ & LF_{4,-1} [m_\Phi] + \frac{1}{4} g_1 g_2^3 LF_{2,2,-1} [m_2, \tilde{\mu}] + g_1 m_2 s_Y \tilde{\mu} c_Y g_2^3 LF_{2,2,0} [m_2, \tilde{\mu}] + \\ & \frac{1}{2} g_1 g_2 \left(c_Y \overline{a_e^{pr}} - s_Y \tilde{\mu} \overline{y_e^{pr}} \right) \left(c_Y a_e^{pr} - s_Y \tilde{\mu} y_e^{pr} \right) LF_{3,1,0} [m_l^p, m_e^r] - \\ & g_1 g_2 \left(c_Y \overline{a_e^{pr}} - s_Y \tilde{\mu} \overline{y_e^{pr}} \right) \left(c_Y a_e^{pr} - s_Y \tilde{\mu} y_e^{pr} \right) LF_{4,1,-1} [m_l^p, m_e^r] + \\ & \frac{1}{2} g_1 g_2 \left(c_Y \overline{a_e^{pr}} - s_Y \tilde{\mu} \overline{y_e^{pr}} \right) \left(c_Y a_e^{pr} - s_Y \tilde{\mu} y_e^{pr} \right) LF_{5,1,-2} [m_l^p, m_e^r] + \\ & \frac{1}{2} g_1 g_2 \left(c_Y \overline{a_d^{pr}} - s_Y \tilde{\mu} \overline{y_d^{pr}} \right) \left(c_Y a_d^{pr} - s_Y \tilde{\mu} y_d^{pr} \right) LF_{3,1,0} [m_q^p, m_d^r] - \\ & 2 g_1 g_2 \left(c_Y \overline{a_d^{pr}} - s_Y \tilde{\mu} \overline{y_d^{pr}} \right) \left(c_Y a_d^{pr} - s_Y \tilde{\mu} y_d^{pr} \right) LF_{4,1,-1} [m_q^p, m_d^r] + \\ & \frac{3}{2} g_1 g_2 \left(c_Y \overline{a_d^{pr}} - s_Y \tilde{\mu} \overline{y_d^{pr}} \right) \left(c_Y a_d^{pr} - s_Y \tilde{\mu} y_d^{pr} \right) LF_{5,1,-2} [m_q^p, m_d^r] + \\ & \frac{1}{4} g_1 g_2 \left(s_Y \overline{a_u^{pr}} - \tilde{\mu} c_Y \overline{y_u^{pr}} \right) \left(s_Y a_u^{pr} - \tilde{\mu} c_Y y_u^{pr} \right) LF_{2,2,0} [m_q^p, m_u^r] + \\ & \frac{1}{4} g_1 g_2 \left(s_Y \overline{a_u^{pr}} - \tilde{\mu} c_Y \overline{y_u^{pr}} \right) \left(s_Y a_u^{pr} - \tilde{\mu} c_Y y_u^{pr} \right) LF_{3,1,0} [m_q^p, m_u^r] - \\ & \frac{1}{4} g_1 g_2 \left(s_Y \overline{a_u^{pr}} - \tilde{\mu} c_Y \overline{y_u^{pr}} \right) \left(s_Y a_u^{pr} - \tilde{\mu} c_Y y_u^{pr} \right) LF_{3,2,-1} [m_q^p, m_u^r] - \\ & \frac{1}{4} g_1 g_2 \left(s_Y \overline{a_u^{pr}} - \tilde{\mu} c_Y \overline{y_u^{pr}} \right) \left(s_Y a_u^{pr} - \tilde{\mu} c_Y y_u^{pr} \right) LF_{4,1,-1} [m_q^p, m_u^r] + \\ & \frac{3}{4} g_1 g_2 \left(s_Y \overline{a_u^{pr}} - \tilde{\mu} c_Y \overline{y_u^{pr}} \right) \left(s_Y a_u^{pr} - \tilde{\mu} c_Y y_u^{pr} \right) LF_{3,1,0} [m_u^r, m_q^p] - \\ & \frac{1}{4} g_1 g_2 \left(s_Y \overline{a_u^{pr}} - \tilde{\mu} c_Y \overline{y_u^{pr}} \right) \left(s_Y a_u^{pr} - \tilde{\mu} c_Y y_u^{pr} \right) LF_{3,2,-1} [m_u^r, m_q^p] - \\ & \frac{9}{4} g_1 g_2 \left(s_Y \overline{a_u^{pr}} - \tilde{\mu} c_Y \overline{y_u^{pr}} \right) \left(s_Y a_u^{pr} - \tilde{\mu} c_Y y_u^{pr} \right) LF_{4,1,-1} [m_u^r, m_q^p] + \\ & \frac{3}{2} g_1 g_2 \left(s_Y \overline{a_u^{pr}} - \tilde{\mu} c_Y \overline{y_u^{pr}} \right) \left(s_Y a_u^{pr} - \tilde{\mu} c_Y y_u^{pr} \right) LF_{5,1,-2} [m_u^r, m_q^p] - \\ & \frac{1}{4} g_2 g_1^3 LF_{3,1,-1} [\tilde{\mu}, m_1] - \frac{1}{2} g_2 m_1 s_Y \tilde{\mu} c_Y g_1^3 LF_{3,1,0} [\tilde{\mu}, m_1] + \frac{3}{4} g_2 g_1^3 LF_{4,1,-2} [\tilde{\mu}, m_1] + \\ & g_2 m_1 s_Y \tilde{\mu} c_Y g_1^3 LF_{4,1,-1} [\tilde{\mu}, m_1] - \frac{1}{2} g_2 g_1^3 LF_{5,1,-3} [\tilde{\mu}, m_1] - g_2 m_1 s_Y \tilde{\mu} c_Y g_1^3 LF_{5,1,-2} [\tilde{\mu}, m_1] - \\ & \frac{7}{4} g_1 g_2^3 LF_{3,1,-1} [\tilde{\mu}, m_2] - \frac{3}{2} g_1 m_2 s_Y \tilde{\mu} c_Y g_2^3 LF_{3,1,0} [\tilde{\mu}, m_2] + \frac{13}{4} g_1 g_2^3 LF_{4,1,-2} [\tilde{\mu}, m_2] + \\ & 5 g_1 m_2 s_Y \tilde{\mu} c_Y g_2^3 LF_{4,1,-1} [\tilde{\mu}, m_2] - \frac{3}{2} g_1 g_2^3 LF_{5,1,-3} [\tilde{\mu}, m_2] - 3 g_1 m_2 s_Y \tilde{\mu} c_Y g_2^3 LF_{5,1,-2} [\tilde{\mu}, m_2] \Big) \end{aligned}$$